

MediTrace

A longitudinal medical tracking tool
for patient empowerment and pharmacovigilance

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<https://caracole.github.io/seguimiento-medico/>

2026 · GNU GPL v3 Licence

Abstract

This paper presents MediTrace, an open-source longitudinal medical tracking tool designed to centralise, visualise and analyse a patient's medical history from heterogeneous PDF documents. Applied to a real case covering the period 2016–2025 (29 documents, 9 analysis dates), the tool identified a critical unreported drug interaction (Haloperidol + Donepezil → double QTc prolongation), a chronic untreated vitamin D deficiency, and insufficient renal monitoring under Dabigatran. The implementation is a Progressive Web Application (PWA) in pure HTML5/JavaScript, running 100% locally with no server, distributed under the GNU GPL v3 licence.

Keywords: longitudinal medical record, pharmacovigilance, drug interaction, health data visualisation, patient empowerment, progressive web application, open source, CIMA/AEMPS.

1. Introduction

Continuity of care is one of the major challenges of modern healthcare systems. When a patient is simultaneously monitored by several specialists in different institutions, their medical data remains fragmented and frequently inaccessible at the time of consultations. Each practitioner starts from scratch, previous test results are not always known, and drug interactions between prescriptions from different departments are not detected.

This problem takes on a critical dimension for elderly polypharmacy patients. An initial diagnostic error — such as the case documented in this work, where a vertebrobasilar stroke was diagnosed 27 hours after the first call to emergency services, the patient having been initially labelled as alcoholism — can have lasting consequences on the entire care pathway.

1.1 Context and motivation

MediTrace was born from a personal initiative: faced with a real medical record scattered across several Spanish institutions (Hospital General de Castellón, Vithas Castellón, Centro de Salud Atzeneta del Maestrat), the need arose to build a tool that centralises, visualises and analyses medical history in its continuity.

The tool's objective is threefold: to enable the patient and their family to understand the evolution of health status; to facilitate dialogue with healthcare professionals by providing a synthetic documentary overview; and to identify clinical signals that the dispersion of data would have masked.

1.2 Original contribution

The main contributions of this work are:

- A system for extracting and structuring biological data from heterogeneous PDFs
- A longitudinal visualisation module for 25+ biological parameters with reference zones
- A clinical correlation module crossing biological parameters, medications and therapeutic adherence
- A pharmacovigilance module based on official CIMA/AEMPS technical data sheets
- Detection of a critical drug interaction (Haloperidol + Donepezil) not reported in the clinical context
- A 100% local implementation, without a server, respecting data confidentiality

2. MediTrace system description

2.1 Technical architecture

MediTrace is implemented as a Progressive Web Application (PWA) in pure HTML5/JavaScript, with no dependency on external cryptographic or medical libraries. The application uses Chart.js 4.4 for visualisations and runs entirely client-side: no data transits through a server. Source PDFs remain on the user's device.

The application comprises seven modules: Summary (KPIs), Charts (25+ parameters), Table (sortable synthesis), Timeline (clinical events + PDF viewer), Correlations (overlaid charts + adherence), Pharmacovigilance (CIMA/AEMPS) and Medical History (structured record).

2.2 Data extraction and structuring

Two main document formats have been processed: structured laboratory reports (Hospital General de Castellón, Vithas) with direct extraction, and CCEE progress notes with values scattered in narrative text (semi-manual extraction). For pre-2022 paper documents, the planned processing chain is: digitisation → OCR via Tesseract (-l spa) → value extraction.

Parameter	Unit	Reference	Available dates
Haemoglobin	g/dL	12.0–18.0	01/2022 → 03/2025 (7 points)
Blood glucose	mg/dL	70–110	01/2022 → 03/2025 (7 points)

			points)
Total cholesterol	mg/dL	—	09/2022 → 02/2025 (4 points)
Creatinine	mg/dL	0.51–0.95	01/2022 → 03/2025 (7 points)
Glomerular filtration	ml/min	60–200	01/2022 → 03/2025 (7 points)
Vitamin D	ng/mL	30–100	09/2022 → 02/2025 (3 points)
Troponin	ng/L	0–14	01/2022 (TEP episode)
NT-ProBNP	pg/mL	20–125	01/2022 (TEP episode)

3. Clinical correlations identified

3.1 Chronic vitamin D deficiency and cognitive decline

Vitamin D was measured on three occasions: 27.10 ng/mL (09/2022), 15.96 ng/mL (04/2024) and 12.33 ng/mL (02/2025), all below the lower reference limit (30 ng/mL) and in progressive decline. The MMSE simultaneously evolves from 25/30 (11/2021) to 22/30 (04/2022) and 16/30 (10/2024). Medical literature establishes an association between vitamin D deficiency and acceleration of cognitive decline [Annweiler et al., 2012]. No substitution treatment is documented.

3.2 Rising cholesterol despite Atorvastatin

Despite atorvastatin 20mg prescribed since before 2022, total cholesterol follows an increasing trajectory: 230 mg/dL (09/2022) → 281.7 mg/dL (04/2024) → 271 mg/dL (02/2025). Calculated LDL reaches 202 mg/dL (target <116 mg/dL). Two non-exclusive explanations: uncertain therapeutic adherence, or insufficient dose (20mg being the minimum dose).

3.3 Renal function and Dabigatran

The CKD-EPI glomerular filtration rate follows a downward trend: 83 (01/2022) → 75.97 (02/2025) → 70.8 (03/2025) → 62.45 (04/2024). The AEMPS information note MUH(FV) 21/2011 recommends assessing renal function at least once a year in patients over 75 years of age treated with dabigatran. Dabigatran is contraindicated with GFR < 30 ml/min.

3.4 Therapeutic adherence

Documentary analysis reveals several confirmed non-adherence episodes: Rivaroxaban refused as non-reimbursed; Dabigatran not taken according to husband's declaration (02/2025); Donepezil temporarily suspended by the physician (04/2022); Memantine probably abandoned after worsening of behaviour; Heparin interrupted by the patient due to bruising.

4. Pharmacovigilance — Interactions and contraindications

4.1 Methodology

The pharmacovigilance analysis was conducted by systematically consulting the official technical data sheets available on the CIMA database (Centro de Información online de Medicamentos de la AEMPS, cima.aemps.es) for each prescribed medication.

4.2 Critical interaction: Haloperidol + Donepezil → double QTc prolongation

⚠ **CRITICAL INTERACTION** — Both medications independently prolong the QTc interval. Their combination may cause serious ventricular arrhythmias (Torsade de pointes), syncope or sudden death. The donepezil technical data sheet (CIMA) explicitly lists 'other antipsychotics' as medications to avoid. The haloperidol technical data sheet (CIMA) warns of increased mortality in elderly patients with dementia. Relative contraindication in a patient with pre-existing moderate tricuspid insufficiency.

4.3 Other important interactions

⚠ **HIGH RISK** — Dabigatran + Aspirin (ASA): the Pradaxa technical data sheet (CIMA) explicitly states that concomitant use of ASA antiplatelet agents 'approximately doubles the rates of major bleeding episodes'. Risk particularly high in patients ≥ 75 years old with the 150 mg/12h dose.

Medication	Interaction	With	Level	CIMA Source
Haloperidol	QTc prolongation	Donepezil	🔴 CRITICAL	FT Haloperidol Esteve
Donepezil	QTc prolongation	Haloperidol	🔴 CRITICAL	FT Aricept / Donepezil
Dabigatran	Double bleeding risk	ASA (Aspirin)	🔴 High	FT Pradaxa 150mg
Dabigatran	Accumulation if GFR↓	Renal function	🔴 Monitoring	AEMPS MUH 21/2011
Atorvastatin	Potential hyperglycaemia	Metabolic profile	🟡 Monitor	FT Atorvastatin Normon
Donepezil + Memantine	No relevant interaction	—	🟢 Safe	FT Domex / TecniGen

5. Implementation

MediTrace is a Progressive Web Application (PWA) in pure HTML5/JavaScript, without any framework or server dependency. Chart.js 4.4 provides the visualisations. Medical data is anonymised in the code (name reduced to initials, SIP number masked) before any public deposit. Source PDFs are never sent to a server: the display operation uses `URL.createObjectURL()` client-side.

The application is available at:

<https://caracole.github.io/seguimiento-medico/>

6. Future work and perspectives

- Automatic extraction of biological values via OCR (Tesseract) and PDF analysis
- Integration of the CIMA API for automatic pharmacovigilance updates
- Automatic alert module for critical thresholds
- Multi-patient support for use in family medicine
- PDF export of the synthesis report
- Formal validation by healthcare professionals

7. Conclusion

This paper has presented MediTrace, an open-source longitudinal medical tracking tool. Applied to a real case (2016–2025), it identified a critical drug interaction (Haloperidol + Donepezil, risk of QTc prolongation), a chronic untreated vitamin D deficiency, and insufficient renal monitoring under Dabigatran.

These results illustrate the added value of a longitudinal, integrated view of medical history. The open-source implementation under the GNU GPL v3 licence, running 100% locally without a server, guarantees data confidentiality and accessibility for all. The author invites the medical and computer science community to evaluate, critique and contribute to this tool.

References

- [1] AEMPS. Information note MUH(FV) 21/2011 — Dabigatran (Pradaxa®) and haemorrhage risk. 2011.
- [2] CIMA AEMPS. Technical Data Sheet Pradaxa 150mg. <https://cima.aemps.es>, 2024.
- [3] CIMA AEMPS. Technical Data Sheet Haloperidol Esteve 5mg/ml. <https://cima.aemps.es>, 2024.
- [4] CIMA AEMPS. Technical Data Sheet Aricept Flas 5mg / Donepezil. <https://cima.aemps.es>, 2024.
- [5] CIMA AEMPS. Technical Data Sheet Domex 10mg/20mg (Donepezil/Memantine). <https://cima.aemps.es>, 2024.
- [6] Annweiler C. et al. Higher vitamin D dietary intake is associated with lower risk of Alzheimer's disease. *Journal of Gerontology*, 2012.
- [7] Giraud, P.-H. MediTrace — Source code. <https://github.com/caracole/seguimiento-medico>, 2026.
- [8] Anthropic. Claude AI Assistant. <https://claude.ai>, 2026.

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